ALGORITHMS II (CSC711) · MINI PROJECT REPORT

Smart Delivery Network on Kalyani

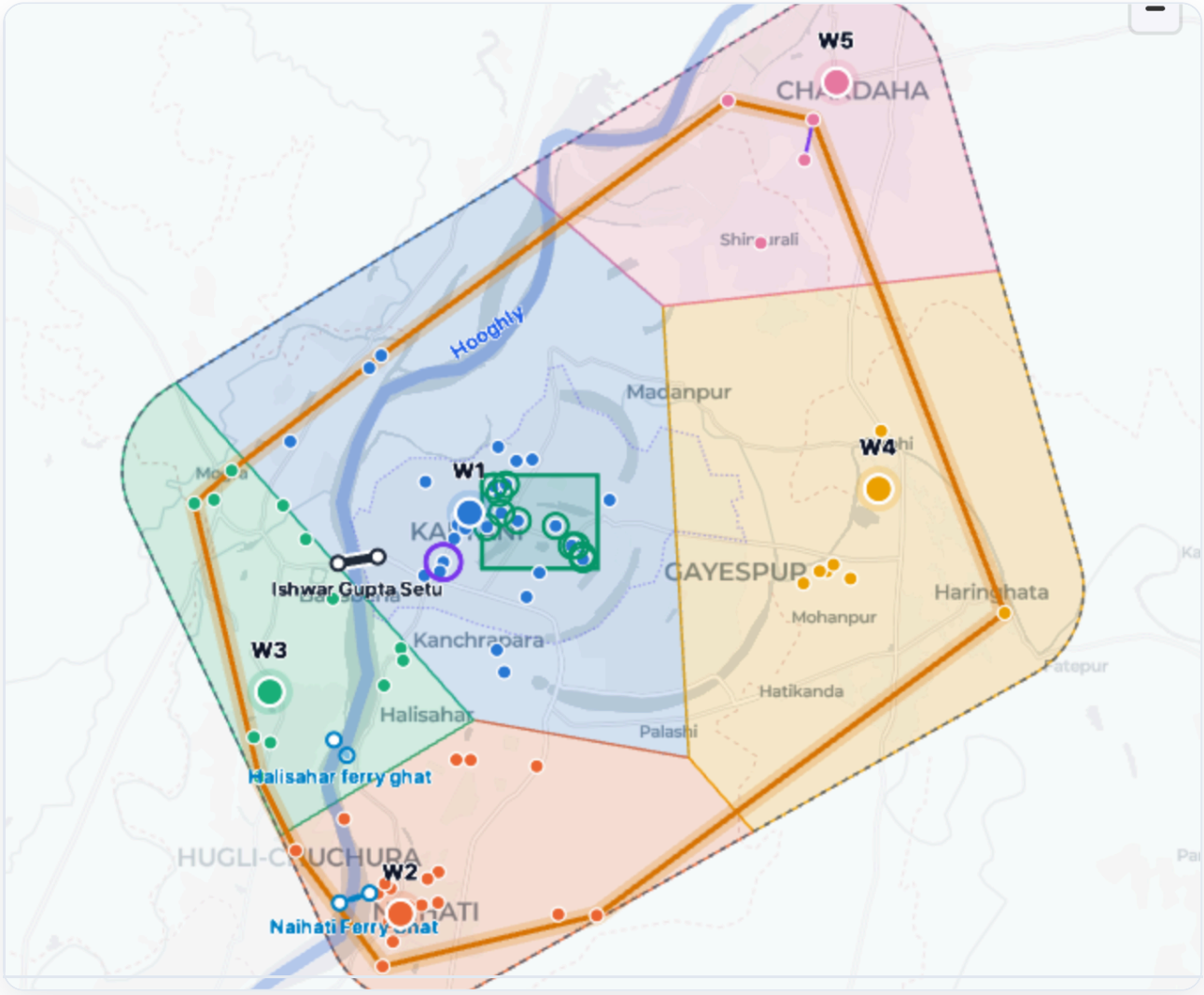
Computational geometry and approximation algorithms for delivery planning on real OpenStreetMap data, on both banks of the Hooghly

Voronoi · KD-tree · Closest pair · Convex hull · Set cover · River check

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SOURCE CODE
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Contents

1 Problem Statement

2 Input Model

3 Algorithms Used

4 How the Algorithms Combine

5 Dashboard

6 Worked Example (default network)

7 Crossing the Hooghly, a Real-World Issue

8 Testing

9 Discussion

10 Data Sources and Attribution

11 Conclusion

A Running the Project (appendix)

Summary. A delivery network around Kalyani, West Bengal, built from 571 real OpenStreetMap places in a 401 km² area on both banks of the Hooghly. Five Algorithms II techniques plan it, namely Voronoi zones (who serves whom), a KD-tree (what is where), divide-and-conquer closest pair (what is near), the convex hull (how far it reaches) and greedy set cover (which sites to keep), shown in an interactive Streamlit dashboard with step-by-step walkthroughs. A river check then measures where straight-line planning breaks. With one road bridge along 27.5 km of river, a van's real trip to a cross-river hub is more than twice the straight line.

1. Problem Statement

A delivery company runs several warehouses in and around **Kalyani, West Bengal**, and serves customer stops from Chakdaha in the north to Naihati in the south, on both banks of the Hooghly river. When the list of stops grows too large to inspect by eye, four questions become hard to answer by hand.

1. Which warehouse should serve which customers?
2. Which stops are close enough to be worth grouping?
3. Which customers lie inside a given area, such as a planned delivery zone?
4. How far does the network reach, and which sites are worth keeping?

Comparing every point with every other point takes quadratic time, which becomes impractical as the number of stops grows. Each question is a standard computational-geometry problem, so each one is handed to the algorithm built for it. The results feed into a single dashboard drawn on the real street map.

All five algorithms measure distance in straight lines. In this region that assumption has a visible real-world flaw, the Hooghly river. A customer on one bank may find that their nearest warehouse "as the crow flies" is on the other bank, and that warehouse can only reach them over a road bridge or by ferry. A sixth, smaller check measures where and by how much the straight-line answers go wrong (§7).

This is a **geometric decision-support tool**, not a full vehicle-routing system. It does not plan turn-by-turn routes, model traffic or fuel, or schedule drivers.

2. Input Model

Every warehouse and customer is a point `Point(id, x, y, kind)` with `kind ∈ {warehouse, customer}`. The algorithms only read `x` and `y`, which are kilometres on a flat local grid. Each point also stores the real place behind it (`name`, `lat`, `lon`, OSM `category`) for display, along with a short label such as `W1` or `C12`.

The study area. The area is the convex hull of five town centres, Kalyani, Naihati, Chakdaha, Mogra and Haringhata, with a 2.5 km buffer. It covers **401 km²**, about 25 × 26 km. Each town's coordinates come from an OpenStreetMap (OSM) node, except Kalyani's, which use the centroid of the *Kalyani City* outline (OSM way 1094723487).

The Hooghly divides the area into two banks.

- **West bank (80 km²).** Mogra, Bansberia and Hugli-Chuchura.
- **East bank (321 km²).** Everything else, including Kalyani, Naihati, Halisahar, Kanchrapara, Haringhata and Chakdaha. Chakdaha lies on the east bank because the river curves around Kalyani's north.

The river's main channel runs for 27.5 km through the area.

The points are real places. The October 2026 extract contains **571 usable named places** inside the area, such as shops, banks, schools, clinics, offices, and college and university buildings. Of these, 128 are on the west bank and 443 on the east bank. In total 52 are tagged as warehouse candidates, such as markets, supermarkets, post offices, warehouses and industrial land.

Latitude and longitude are projected to a flat km grid. An equirectangular projection is anchored at the study area's south-west corner (lat_0 = 22.864°, lon_0 = 88.349°).

```
x = (lon - lon₀) · 111.320 · cos(lat₀)     y = (lat - lat₀) · 110.574
```

So 1 unit is 1 km, and every distance can be reported in metres. The grid stays accurate at this scale. Across all 157,880 pairs of places at least 0.5 km apart, the flat-grid distance stays within 0.2 % of the true distance on the WGS84 ellipsoid, with a worst case of 0.15 %. The unit tests compare a 9 km diagonal with the haversine great-circle distance and check that it meets the same tolerance.

Sampling is seeded. `sample_network(region, m, n, seed)` uses Python's `random.Random(seed)` to pick the network, so the same seed always gives the same network.

- **Warehouses** are drawn from the 52 candidates first, then from any place. They must be at least `0.55 · √(area / m)` km apart (4.92 km for m = 5). This stops two hubs from sitting next to each other and avoids degenerate service zones. If the area runs out of far-apart places, the minimum gap shrinks by 20 % and the search repeats.
- **Customers** are a uniform random sample of the remaining places. Places at exactly the same spot, to about 1 m, are skipped, so no two customers coincide.

The defaults are seed 42, 5 warehouses and 80 customers. The sidebar allows 2 to 8 warehouses and 10 to 300 customers. The limit of 8 lets each zone keep its own colour from a fixed palette that was checked for colour blindness.

CSV upload. Users can also upload any network as a CSV with the columns `type, lat, lon` and an optional `name`, where `type` is `warehouse` or `customer`. A file needs at least 1 warehouse and 2 customers, and at most 8 warehouses. The flat grid is then anchored at the uploaded points' own south-west corner, so the tool works for any city, and the zones are clipped to a buffered convex hull of the points. Uploaded networks have no river, so the river check is skipped for them. The Data page exports the current network as a CSV with these columns plus `label` and `category`, so a downloaded file can be edited and uploaded again.

2.1 Data pipeline

The OSM extract is downloaded once by `scripts/fetch_osm.py` and saved in the project. The algorithms therefore run offline, and only the background street-map tiles need internet access.

1. **Define the area.** The script builds the padded convex hull of the five anchor towns.
2. **Fetch.** Queries go to the Overpass API for the following data.
   - Every named feature inside the area, plus warehouses and industrial land
   - The ways of the OSM river relation "Hooghly" near the area
   - The road bridges, ferry routes and ferry terminals near the area
   - The Kalyani City outline, which is drawn on the map for context

   Overpass is often overloaded, so the script tries several mirrors and retries with exponential back-off. It downloads only the raw files that are missing, and the `--refresh` option downloads them all again.
3. **Keep the raw responses.** The untouched JSON is saved in `data/2026-10/raw/`. Each month has its own folder, and older folders are never edited. The files from the first, Kalyani-City-only extract remain in the same folder as legacy data.
4. **Clean the places.** Each feature is classified by its main OSM tag, such as amenity, shop, office or healthcare. Roads, rivers, rail lines, parks, parking and plain residential land are skipped. Duplicates are also removed when a place is mapped both as a node and as a building. Warehouse candidates are flagged by tag, and city and town nodes become map labels (10 towns).

5. **Build the river.** The relation's pieces are joined into one main channel. Side channels that loop around islands are dropped, so the line never crosses itself.
6. **Build the crossings (3).**
   - **Ishwar Gupta Setu.** The road bridge between Kalyani and Bansberia, 1.04 km long.
   - **Naihati Ferry Ghat.** Two OSM ferry terminals on opposite banks, paired, 0.82 km apart.
   - **Halisahar ferry ghat.** Only one terminal is mapped, so its far landing is estimated by mirroring it across the channel, 0.52 km away.

   Rail-only bridges (Jubilee and Sampreeti, Naihati–Bandel) are left out because delivery vehicles cannot use them. Ghats the team knows locally can be added by hand in `data/manual/ferries.csv`.
7. **Save.** The outputs are `region.geojson`, `region_places.csv`, `towns.csv`, `river.geojson` and `crossings.csv`. The app always loads the newest complete month.

3. Algorithms Used

#	Algorithm	Question answered	Technique	Complexity
1	Convex hull (Andrew's monotone chain, a Graham-scan variant)	How far does the network reach?	Sort the points by (x, y), then build the lower and upper chains, popping any point that makes a non-left turn	O(n log n)
2	Closest pair (divide and conquer), region-wide and per zone	Which two stops are nearest each other?	Split at the x-median, recurse on both halves, then scan only the thin strip around the dividing line for a closer cross-pair	O(n log n), against $O(n^2)$ for brute force
3	KD-tree + range search	Which customers lie inside this rectangle?	Build a binary tree over the customers with alternating x and y median splits. A range query skips whole subtrees that cannot meet the query box	Build O(n log n), query $O(\sqrt{n} + k)$
4	Voronoi diagram	Which warehouse is nearest to each part of the area?	Cells come from scipy/Qhull and are clipped to the study area with shapely. Each customer is assigned by a direct nearest-site test	Cells O(m log m) expected, assignment O(m) per customer
5	Greedy set cover (optional extension)	Which few warehouses cover everyone?	Each warehouse covers the customers within a radius. Repeatedly keep the warehouse that covers the most still-uncovered customers	The exact optimum is NP-hard, and greedy stays within $H(n) \approx \ln n$ of it
+	River check (real-world extension, §7)	Where does straight-line planning break?	Find which bank each point is on. Trips across the river go via the best allowed bridge or ferry	O(c) per trip, $O(n \cdot m \cdot c)$ for the whole check

Here n is the number of customers, m the number of warehouses, k the number of points a query reports and c the number of river crossings.

3.1 Convex Hull with Andrew's Monotone Chain

The hull uses **Andrew's monotone chain**, a variant of Graham's scan. Graham's scan sorts the points by polar angle around the lowest point. Andrew's version sorts them by `(x, y)` instead and builds the hull in two halves, which avoids angle comparisons and handles ties simply.

The customers are sorted by `(x, y)`, with duplicates removed. The **lower chain** is built by scanning them from west to east. Each new point is pushed onto a stack, but first the top of the stack is popped for as long as the last two stack points and the new point fail to make a left (counter-clockwise) turn. The turn is tested with the sign of a cross product. The **upper chain** is built the same way, from east to west. Joining the two chains gives the hull in counter-clockwise order, starting from the westmost stop. Collinear boundary points are dropped. Each point is pushed and popped at most once per chain, so the scan takes linear time after the O(n log n) sort.

The area the hull encloses is reported as the network's **reach**.

3.2 Closest Pair by Divide and Conquer

The customers are sorted by x and split at the median into two halves. Each half is solved recursively, and three or fewer points are solved by brute force. Let `δ` be the smaller of the best distances in the two halves.

Any closer pair must have one point on each side, with both points within `δ` of the dividing vertical line. So only the strip of width 2δ around that line needs to be checked again. The strip's points are taken in y order, and each point is compared only with the points above it until the y-gap reaches `δ`. A packing argument limits this to a constant number of comparisons per point.

No level sorts anything. The points are sorted by x once at the start, and each recursive call returns its points sorted by y. That order is built by merging the two halves' y-orders in linear time, as in merge sort. The strip is filtered from the merged list, which gives T(n) = 2T(n/2) + O(n) = O(n log n). Comparing every pair would take $O(n^2)$.

The dashboard runs the algorithm on all customers to find the two nearest stops in the whole region. It also runs it **inside each Voronoi zone**, giving each warehouse its own pair of stops worth grouping into one delivery.

3.3 KD-Tree and Range Search

A KD-tree is built by recursively splitting the customer set at the median x (at even depth) or the median y (at odd depth). This keeps the tree balanced, with a depth of about $\log_2 n$. The points are sorted once by x and once by y. Each node takes the median of the order for its axis and splits both orders in one linear pass by comparing ranks with the median's rank. Both orders stay sorted without any re-sorting, which gives T(n) = 2T(n/2) + O(n) = O(n log n).

A rectangle range query starts at the root and recurses down the tree. At each node it reports the node's point if it lies inside the box, then visits a child only if the box reaches that side of the splitting line. Whole subtrees are skipped when the box cannot touch their part of the area. A query therefore visits only the points near the box and the ones it reports. For a balanced 2D tree this costs $O(\sqrt{n} + k)$, instead of a scan of every customer.

3.4 Voronoi Diagram

Each warehouse owns a cell of the area, which holds the points closer to that warehouse than to any other warehouse.

- **Cells.** The cell geometry comes from `scipy.spatial.Voronoi`, which wraps Qhull. Qhull builds the Delaunay triangulation of the sites as the lower convex hull of the sites lifted onto a paraboloid. It then returns the dual, the Voronoi diagram, in O(m log m) expected time. Four helper sites far outside the area are added so every real warehouse gets a closed cell. Each cell is then clipped to the study area with shapely, so the zones match the map and their areas add up exactly to the area's total.
- **Assignment.** A direct nearest-site test decides which warehouse serves each customer. The customer is compared with every warehouse, and the nearest one wins. This takes O(m) per customer and is exact by the definition of a Voronoi cell, so the decision never depends on polygon geometry. The tests check that every customer falls inside its assigned warehouse's clipped cell.

3.5 Greedy Set Cover

The warehouses are the candidate sites, and each one covers the customers within a chosen coverage radius, measured in straight-line distance. Finding the true minimum number of sites that covers every customer is an NP-hard set-cover problem, so a greedy heuristic is used instead. It repeatedly keeps the candidate that covers the most still-uncovered customers, until every customer is covered or no candidate can add anyone new. This greedy rule is proven to stay within a factor `H(n) = 1 + 1/2 + ... + 1/n ≈ ln n` of the true optimum. The set-cover example from the project overview, with four sites, nine customers and the answer W2 then W4, is one of the unit tests.

4. How the Algorithms Combine

- **Voronoi + KD-tree.** Voronoi decides which warehouse owns each customer. The KD-tree then pulls out the customers inside any rectangle on the map, and their warehouse labels come along with them. One query therefore answers both "who is here" and "who serves them".
- **Voronoi + closest pair.** Closest pair runs inside each warehouse's zone to find the stops worth grouping into a shared delivery. This is proximity analysis only, and it does not solve routing.
- **Convex hull + everything.** The hull marks the outer edge of the customer set and shows how far the network reaches compared with the whole area. Its corners are the outlying stops, which are also the stops most likely to fall outside a coverage radius.
- **Set cover + Voronoi.** Voronoi assigns each customer to exactly one warehouse. Set cover asks a different question, which warehouses are needed at all for a given reach.
- **Voronoi + river check.** The river check takes Voronoi's straight-line assignment and tests it against the real geography. Wherever a zone spills across the Hooghly, it measures the real trip and finds the hub that is actually nearest.

5. Dashboard

The dashboard is a Streamlit app (`app.py` and `ui/`) with six pages in a top navigation bar. The maps use deck.gl through pydeck on CARTO's light "Positron" basemap, which is built from OpenStreetMap data. The app uses a single light theme. Every page has a header with the project name, the course and summary chips, and the Network map and How it works pages also list the team.

The **sidebar** chooses the data source (*Kalyani region* or *Upload CSV*), the number of warehouses (2 to 8) and customers (10 to 300), and the random seed. A *Shuffle network* button picks a new seed. The sidebar also shows the course, the team and the OSM and CARTO credits.

1. **Network map.** The headline cards show the following.
    - Service zones and the busiest zone
    - Delivery stops and their average distance to their hub
    - Network reach, with the hull area and the number of boundary stops
    - The closest pair
    - The customers in the search area
    - The customers whose straight-line hub lies across the Hooghly

    The map shows the Voronoi zones, coloured by warehouse and clipped to the study area. It also shows the convex hull, the KD-tree search area with its hits ringed, the closest pair across the region as a thick line and the per-zone pairs as thin lines. The river with its bridge and ferries and the Kalyani City outline add context, and each layer can be switched off. Clicking any dot shows details about that place. A warehouse shows how many customers it serves, and a customer shows its three nearest warehouses with their distances. Either can become the centre of the search area.

    A side panel has three tabs.

    - **Search area.** The centre can be Kalyani or any place. Width and height range from 0.5 to 10 km, with a default of 3.0 × 2.4 km. Hits are listed from nearest to farthest.

- **Zones.** The number of customers and the average and farthest distance for each zone.
- **Closest stops.** The closest pair across the region and in each zone.

2. **River crossing.** This page runs the river check of §7. A switch chooses between *Delivery van*, which may use road bridges only, and *Two-wheeler*, which may use bridges and ferries. The cards show the following.
   - The west-bank customers
   - How many have their hub across the river
   - The average straight-line and real trip
   - The extra distance
   - How many have a nearer hub once the river is taken into account

   The map draws each affected customer's straight line next to its real route over the bridge or ferry. A side list ranks the affected customers by extra distance and lists the three crossings.
3. **Coverage planner.** Greedy set cover with a radius slider from 1 to 10 km and a default of 4 km. The cards show the following.
   - How many warehouses are kept
   - The share of customers covered
   - How many are not covered
   - The number of rounds
   - The H(n) guarantee

   The map draws each kept warehouse's radius in its zone colour, shows the warehouses that could close as grey outlines, and rings every uncovered customer in red. A round-by-round timeline lists each pick, how many new customers it covered and the running total.
4. **Walkthrough.** This page steps through every decision of five algorithms on the customers currently on the map.
   - **Convex hull.** The chain being built and each popped point.
   - **Closest pair.** Each split line, base case and strip of width 2δ.
   - **KD-tree build.** Each median split.
   - **KD-tree search.** Each visited node and skipped subtree.
   - **Voronoi assignment.** One customer compared with each warehouse in turn.

   Arrows and a slider move between steps. The traces come from instrumented copies of the algorithms in `src/traces.py`. These copies are kept apart from the tested core code so the teaching output cannot change a result, and a test checks that the traces agree with the core algorithms.
5. **Data.** The full point set as a table with label, name, type, OSM category, zone, bank, distance to hub, x and y in km, and latitude and longitude. The page also has a CSV download, a bar chart of customers per zone and a chart of the kinds of places.
6. **How it works.** One card per algorithm and one for the river check, each with its question and complexity. The page then explains the OSM pipeline in four steps and sets out the scope and limits.

6. Worked Example (default network)

The input is the October 2026 OSM extract with the app's defaults, **seed 42, 5 warehouses and 80 customers**, the default search area (3.0 × 2.4 km around Kalyani City) and the default 4 km coverage radius. Every number below can be reproduced with `python report/worked_example.py`, which calls the same code as the app.

Warehouses chosen by the seeded sampler (no two are closer than 6.64 km)

Label	OSM place	Category	Bank	x (km)	y (km)
W1	More (Kalyani)	supermarket	east	9.00	12.66
W2	Naihati Post Office	post office	east	7.21	2.35
W3	Industrial site	industrial	west	3.83	8.06
W4	Flipkart Warehouse	warehouse	east	19.60	13.27
W5	Chakdaha Post Office	post office	east	18.51	23.72

Of the 80 customers, 13 are on the west bank.

Results

Step	Input	Output
Voronoi	5 warehouse sites, 80 customers	5 zones clipped to the area, with areas summing to 400.78 km². The zones hold W1 39, W2 18, W3 12, W4 7 and W5 4 customers. The average straight-line distance to the assigned hub is 2.23 km.
Convex hull	80 customers	**10 boundary stops**, enclosing 283.1 km² (71 % of the study area)
Closest pair (region-wide)	80 customers	**C36 Shivam Enterprise ↔ C3 S B Printers, 10 m** (brute force would compare 3,160 pairs)
Closest pair (per zone)	Each zone's customers	W1 (C36 ↔ C3, 10 m) · W4 (C50 APC Ray Lecture Hall Complex ↔ C22 Indian Overseas Bank, 185 m) · W2 (C15 Bankim Pathagar ↔ C25 Katyayani Girls High School, 190 m) · W3 (C28 Pharmacy ↔ C38 Dey's Residence, 318 m) · W5 (C24 Palpara ↔ C13 Bapuji Balika Vidyamandir, 1.06 km)
KD-tree range query	The default search area of 3.0 × 2.4 km centred on Kalyani City's centroid (x 9.32–12.32 km, y 11.22–13.62 km)	**11 customers**, all in zone W1, among them Kalyani Court, the University of Kalyani, Vidyasagar Manch and the A.P.J. Abdul Kalam Auditorium. The search visited 25 of the 80 tree nodes and skipped 8 subtrees, and a brute-force filter returns the same 11.
Greedy set cover	5 candidate warehouses, radius 4 km	**5 rounds**, keeping W1 (+36, 36/80), W2 (+13, 49/80), W3 (+6, 55/80), W4 (+6, 61/80) and W5 (+3, 64/80). No warehouse can close. **16 customers** are more than 4 km from every warehouse, and the farthest, C71 Jagorani Football Ground, is 5.77 km away.

Step-by-step traces. These counts show the work each algorithm does on this input.

- **Convex hull.** 308 steps (160 pushes, 148 pops).
- **Closest pair.** 125 steps (31 divides, 32 base cases, 31 combines and 31 strip checks).

- **KD-tree build.** 80 median splits over 7 levels.
- **KD-tree search.** 33 steps (25 visits, 8 skipped subtrees).
- **Voronoi trace.** Customer C1 (Ramchandrapur Health Centre) is 10.87 km from W1, 5.09 km from W2, 10.24 km from W3, 13.18 km from W4 and 22.31 km from W5, so it joins zone W2.

What the numbers say

- W1 (More, a supermarket in Kalyani) is the busiest hub, with 39 of the 80 customers. Kalyani holds most of the mapped places, and the whole default search area falls inside W1's zone.
- W4's zone is nearly as large as W1's (117 km² against 119 km²) but has only 7 customers. The east of the area, around Gayespur and Haringhata, has few places in OSM. Zone sizes reflect mapping density, not real demand.
- The hull covers 71 % of the study area, so the customers are spread over the whole region rather than only Kalyani.
- At 4 km every warehouse is needed, because each one covers customers that no other warehouse reaches. Even so, 20 % of the customers stay out of reach, among them the hull's corners C71, C76 and C1.
- The closest pair is only 10 m apart. An electrical shop and a print shop share essentially the same spot, so one delivery could serve both.

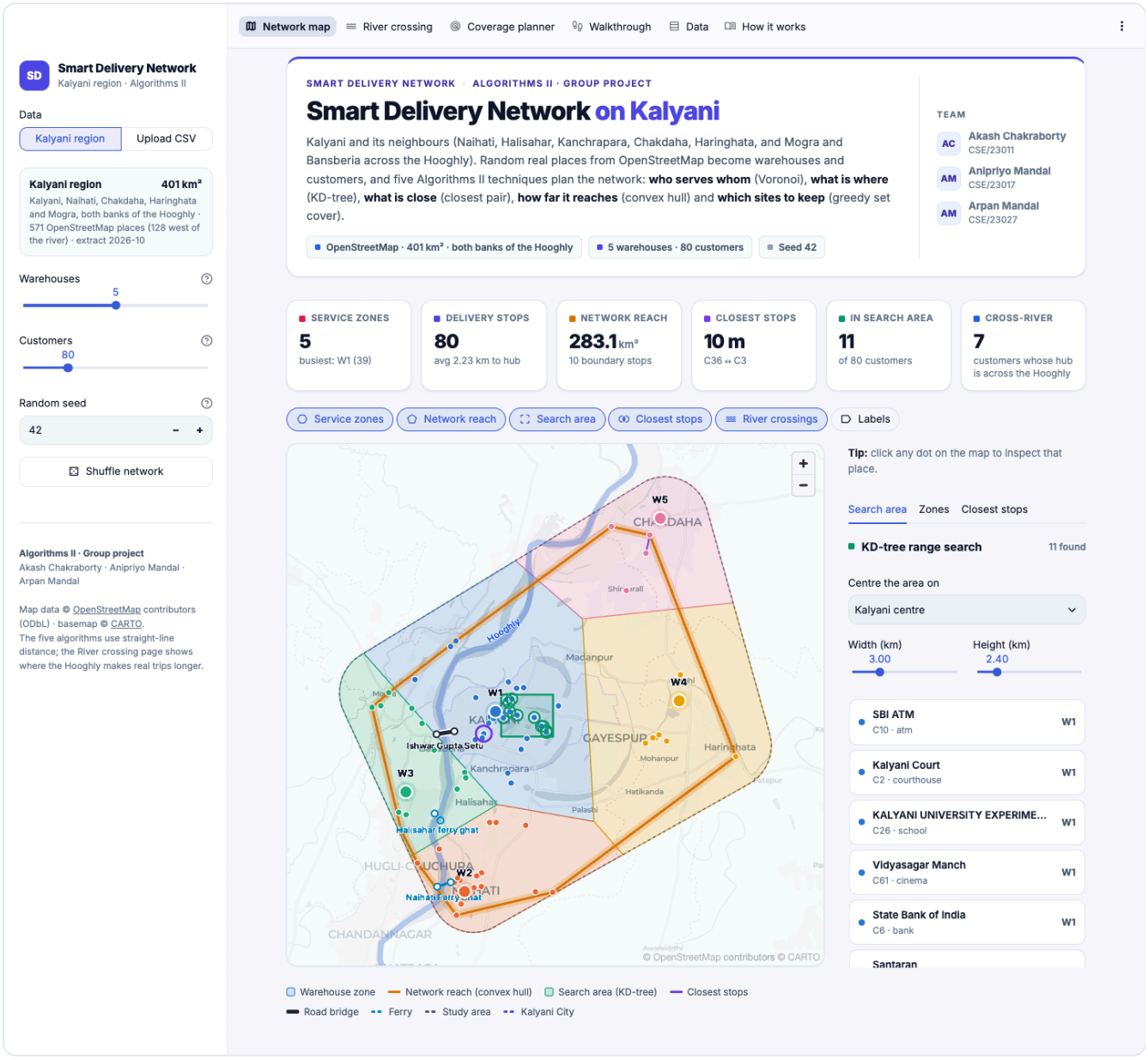


Figure 1. Network map. The Network map page for the default network, with the Voronoi service zones coloured by warehouse and clipped to the study area, the convex hull (orange), the KD-tree search area around Kalyani with its 11 hits ringed (green), the closest stops across the region and per zone (purple), and the Hooghly with Ishwar Gupta Setu and the two ferry ghats. The cards above the map report 5 zones, 80 stops, a 283.1 km² reach with 10 boundary stops, a 10 m closest pair, 11 customers in the search area and 7 customers whose hub is across the river. Map data © OpenStreetMap contributors (ODbL), basemap © CARTO.

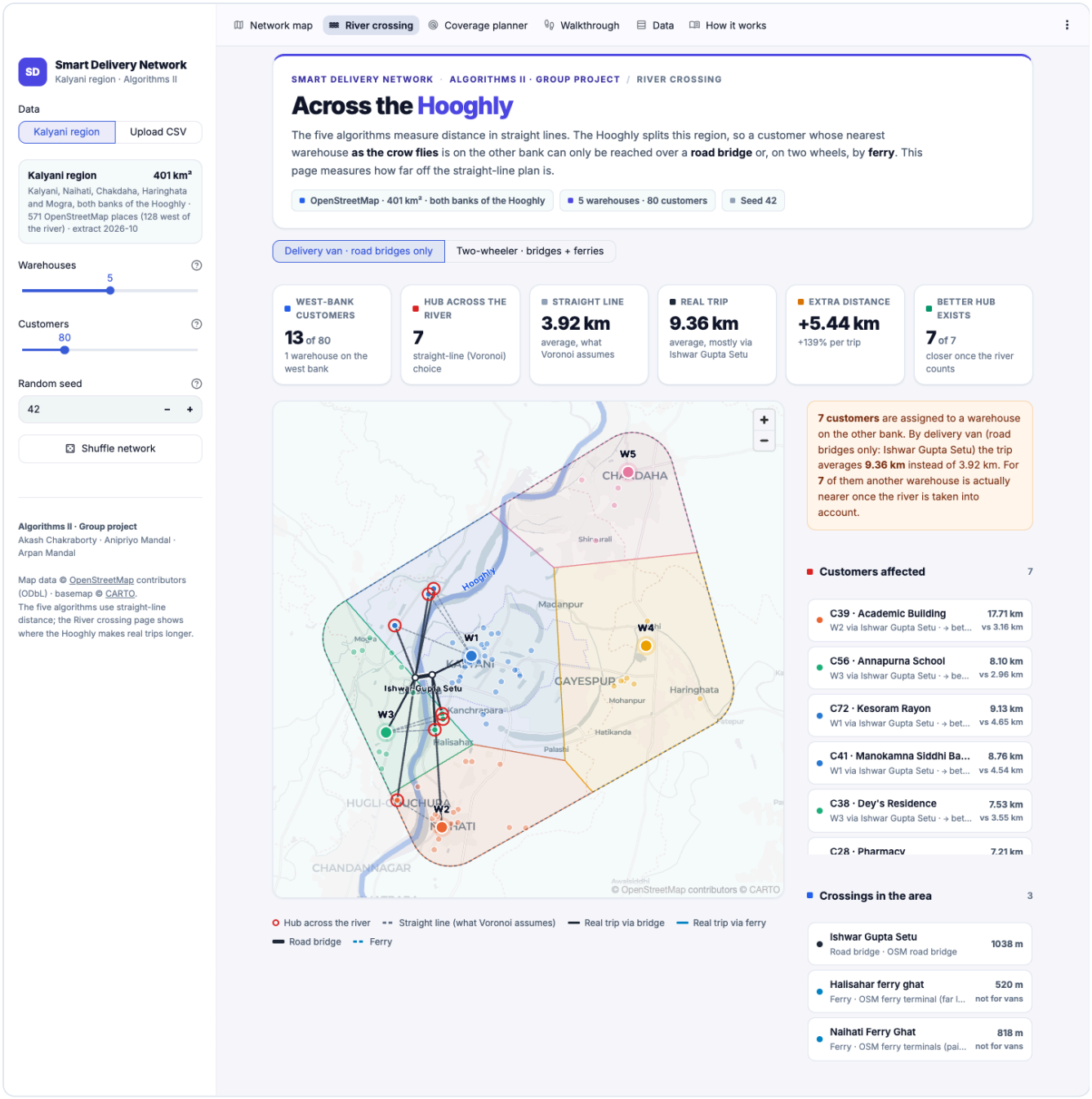


Figure 2. River crossing. The River crossing page in delivery-van mode. The 7 customers whose straight-line hub is across the Hooghly are ringed in red, with their straight lines (dashed) and their real routes over Ishwar Gupta Setu (solid). The cards report 13 west-bank customers, 7 with a hub across the river, an average of 3.92 km in a straight line against 9.36 km in reality (+139 %), and a nearer hub for all 7. Map data © OpenStreetMap contributors (ODbL), basemap © CARTO.

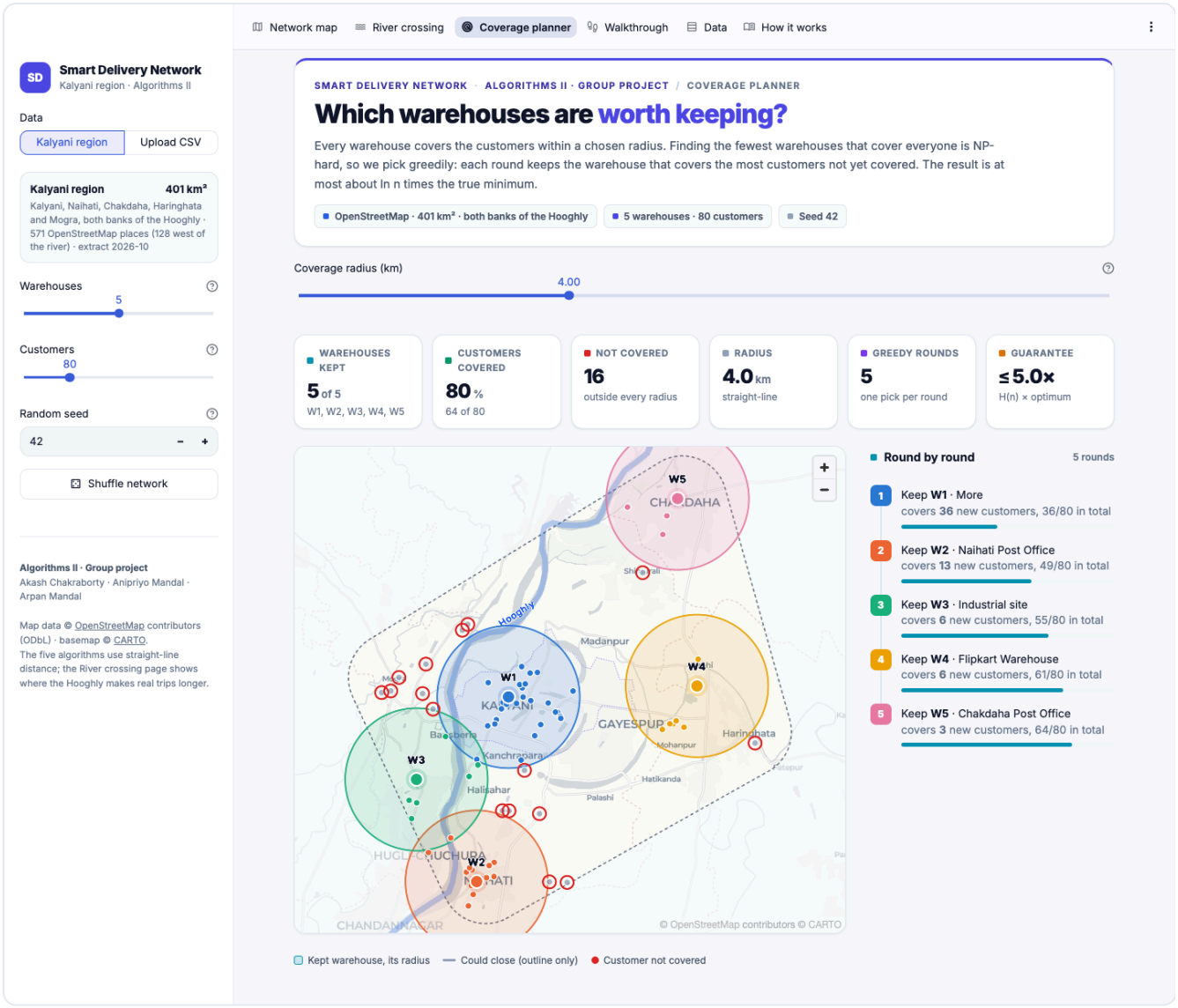


Figure 3. Coverage planner. The Coverage planner at the default 4 km radius, with all five warehouses' radii in their zone colours and the 16 uncovered customers ringed in red. The timeline on the right lists the five greedy rounds (W1, W2, W3, W4, W5) and the running total up to 64 of 80. Map data © OpenStreetMap contributors (ODbL), basemap © CARTO.

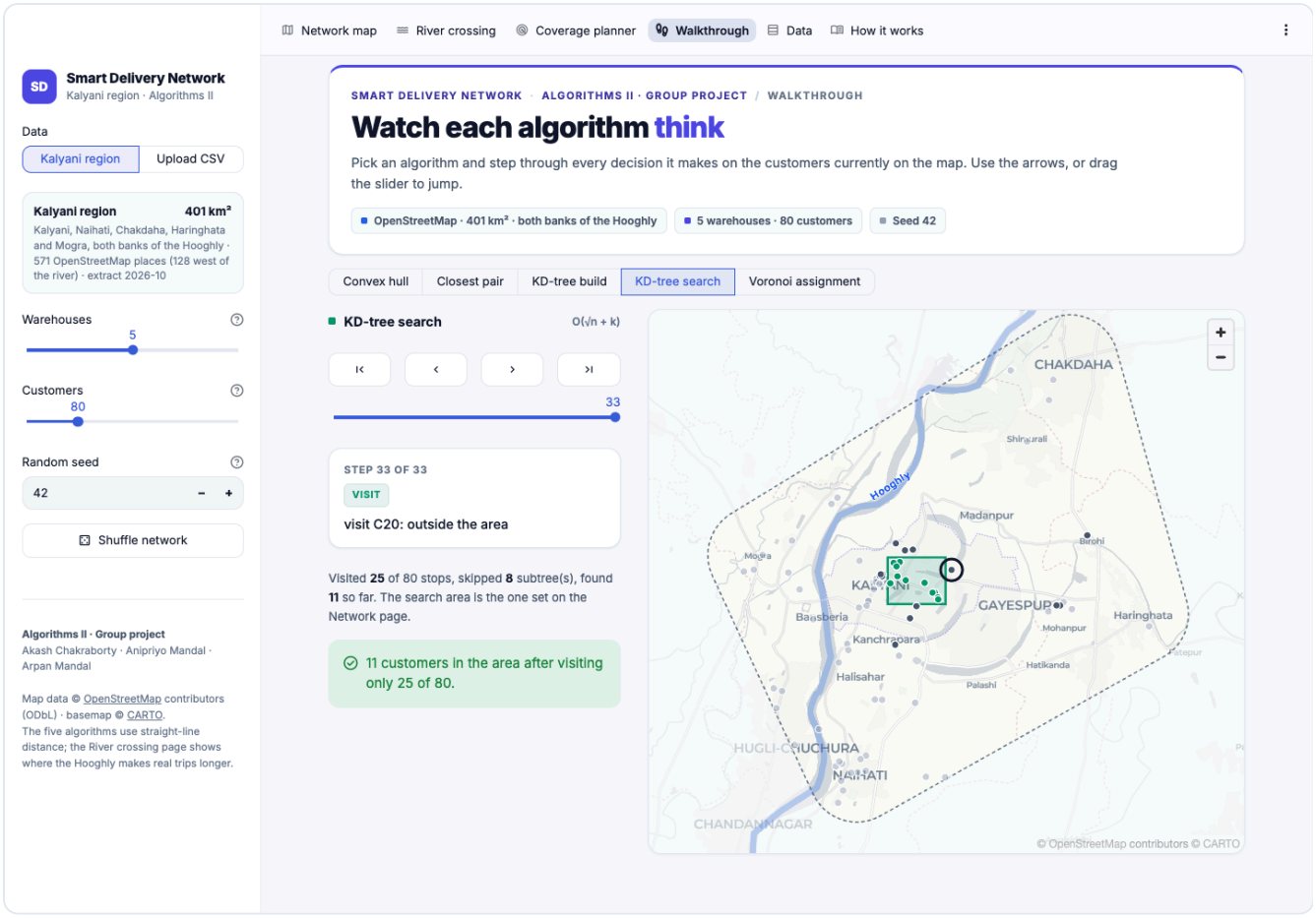


Figure 4. Walkthrough of the KD-tree search. The Walkthrough page at step 33 of 33 of the KD-tree search, with the default search area around Kalyani (green box), the 11 customers found inside it (green), the nodes the search visited (dark grey) and the stops it never had to look at (light grey). The summary reads visited 25 of 80 stops, skipped 8 subtrees, found 11. Map data © OpenStreetMap contributors (ODbL), basemap © CARTO.

7. Crossing the Hooghly, a Real-World Issue

Voronoi assigns each customer to the warehouse that is nearest **in a straight line**. Across a river this can be wrong, because the trip has to use a crossing. The river check in `src/river.py` measures how wrong.

Model

- **Banks.** The river's main channel is extended at both ends past the study area and used to cut the area into two pieces with shapely. The piece whose centroid lies further west is the west bank, and a point's bank is the piece that contains it. Points that fall in neither piece are decided by a cross-product side test against the nearest stretch of the channel.
- **Trips.** A trip between two points on the same bank is the straight line. For points on opposite banks, the trip is the shortest `|p - landing on p's bank| + crossing length + |landing on q's bank - q|` over the allowed crossings, which takes O(c). A **delivery van** may use road bridges only, which here means Ishwar Gupta Setu. A **two-wheeler** may also take the two ferries.
- **Check.** For every customer whose straight-line (Voronoi) hub is on the other bank, the check records the straight-line distance, the real trip and the crossing used. It then finds the hub that is really nearest once the river counts, by routing the customer to every warehouse, which takes O(m · c) per customer.

Results on the default network (13 customers and one warehouse, W3, are on the west bank)

	Delivery van (road bridges only)	Two-wheeler (bridges + ferries)
Customers whose straight-line hub is across the river	7	7
Average straight-line distance (what Voronoi assumes)	3.92 km	3.92 km
Average real trip	9.36 km	6.32 km
Extra distance per trip	**+5.44 km (+139 %)**	**+2.40 km (+61 %)**
Crossings used	Ishwar Gupta Setu × 7	Ishwar Gupta Setu × 3, Halisahar ferry × 3, Naihati ferry × 1
A nearer hub exists once the river counts	7 of 7	5 of 7

- **The worst case is C39 (Academic Building), on the west bank.** Voronoi gives it to W2 in Naihati, 3.16 km away across the water. By van the only route goes up to Ishwar Gupta Setu and back down, 17.71 km in all, while W3 on its own bank is 4.14 km away. By two-wheeler the Naihati ferry cuts the trip to 3.54 km, so W2 is the right hub after all.
- **Some of the "nearer hub" cases are marginal.** For C72 (Kesoram Rayon) and C41 the van trips to the Voronoi hub W1 are 9.13 km and 8.76 km, and W3 is only 10 m and 40 m shorter. The river matters most for customers near the water and far from the only road bridge.

In short, with only one road bridge along the 27.5 km of river inside the area, the straight-line service zones are wrong for about 1 in 11 customers when deliveries go by van. For two-wheelers, the ferries cut the extra distance by more than half.

8. Testing

There are **36 automated tests** in three files, and all of them pass (`python -m pytest tests/ -v`).

`tests/test_algorithms.py` (19 tests) checks each algorithm against a brute-force reference or a known answer.

- Closest pair matches an $O(n^2)$ pairwise scan over five random seeds and many input sizes. It handles duplicate coordinates and passes a small hand-made case. The y-merge is sorted and stable, and a test fails if the recursion ever sorts again after the single initial sort by x, which guards the O(n log n) bound.
- Convex hull keeps only the corners of a square with an interior point, and handles a fully collinear point set as a degenerate case.
- KD-tree range queries match a brute-force filter, also with many duplicate coordinates, and an empty box returns nothing. The tree keeps the split invariant and stays balanced when values tie, and a test fails if the build sorts anything beyond the two initial presorts.
- Greedy set cover covers every coverable customer, reproduces the overview document's worked example (W2 then W4) and stops when no candidate adds anything new.
- The Walkthrough traces agree with the core algorithms for the hull, closest pair and KD-tree search, also on duplicate points, and the Voronoi trace ends at the nearest warehouse.

`tests/test_geo.py` (10 tests) checks the geography and data handling.

- Projecting latitude and longitude to the grid and back gives the original values, and a 9 km diagonal agrees with the haversine distance to within 0.2 %.
- The OSM extract is sensible, with an area of 300–500 km^2, at least 400 places, at least 50 of them on the west bank, and all of them inside the area.
- Sampling is reproducible for a fixed seed. It never puts a customer on a warehouse or two customers on the same spot, and it keeps warehouses more than 2 km apart.
- For 1, 2, 5 and 8 warehouses, the clipped Voronoi zones partition the area, so their areas sum to the area's total, and every customer lies inside the zone of its assigned warehouse (4 tests).
- A network survives a CSV export and re-import, and a CSV with missing columns is rejected.

`tests/test_river.py` (7 tests) checks the river model, mostly on a toy 10 × 10 km square split by a north–south river with one bridge and one ferry.

- The two banks are found, and every crossing is oriented from west to east.
- A trip between two points on the same bank is the straight line.
- A trip across the river uses the best allowed crossing. The two-wheeler takes the ferry, while the van must use the bridge and travels further.
- With no allowed crossing, the trip is impossible (infinite).
- The river check flags only cross-bank assignments and finds the better hub.
- Joining the river pieces gives one simple line past an island.
- On the real data, Mogra is on the west bank, and Kalyani, Naihati, Chakdaha and Haringhata are on the east bank. Ishwar Gupta Setu is a bridge, every crossing joins the two banks, and no real trip is shorter than its straight line.

9. Discussion

Limitations

- Euclidean (straight-line) distance stands in for road distance. Apart from the river check, the model has no roads, one-way streets or traffic.
- The river check is itself simplified.
  - Distances on each bank are still straight lines.
  - The OSM channel geometry is coarse, with vertices up to 1.5 km apart, so places right on the bank can land on the wrong side.
  - The far landing of the Halisahar ferry is estimated.
  - Ferries have no timetables or waiting times.
  - Only three crossings are known in OSM for this stretch of the river.
- The system does not plan routes. It finds the network's structure (who serves whom, who is nearby and how far the network reaches) rather than an ordered path for a vehicle to drive.
- The customers are random real OSM places standing in for customers, not real orders. OSM coverage is uneven, with places clustered in Kalyani and sparse in the east and north, so some zones get few customers, as W4 and W5 do in the example above.
- Greedy set cover gives an approximate rather than optimal warehouse subset. For instances this small, an exact solver would also be feasible.
- The Voronoi cells come from scipy/Qhull, not from Fortune's sweep-line algorithm named in the project overview. Both run in O(m log m), but the sweep itself is neither implemented nor shown in the Walkthrough.
- The flat projection is valid only at regional scale. An uploaded CSV that spans a whole country would distort distances.

Future Scope

- Replace straight-line distance with road-network distance, such as shortest paths over the OSM street graph, which the same Overpass pipeline can download. The river would then be handled automatically.
- Make the assignment river-aware by assigning each customer by real trip length instead of straight-line distance (a Voronoi diagram under a different distance), and let the Coverage planner use the same distance.
- Add the ferry ghats the team knows locally to `data/manual/ferries.csv`.
- Extend the model from static demand, where all customers must always be served, to **capacitated** facility assignment, where each warehouse has a maximum number of customers it can serve and customers are weighted by expected demand.
- Combine the Voronoi and KD-tree layers with an actual routing stage, such as nearest-neighbour or 2-opt TSP heuristics, for each warehouse zone.
- Implement Fortune's sweep for the Voronoi diagram and add it to the Walkthrough.

10. Data Sources and Attribution

- **Map data © OpenStreetMap contributors**, available under the Open Database License (ODbL), see openstreetmap.org/copyright. The places, towns, river, crossings and Kalyani City outline were downloaded through the Overpass API in October 2026 and are stored in `data/2026-10/`.
- **Basemap tiles © CARTO**, see carto.com/attributions. The dashboard's maps use the light "Positron" style, which is drawn from OpenStreetMap data.
- The software used is Streamlit, pydeck/deck.gl, SciPy (Qhull), shapely, pandas, Altair and pytest.

11. Conclusion

Voronoi decides who serves whom, the KD-tree finds what is where, closest pair finds what is near, the convex hull shows how far the network reaches, and greedy set cover picks the sites worth keeping. Run on real OpenStreetMap places around Kalyani and drawn back on the street map, these algorithms turn a list of warehouse and customer coordinates into an interactive, geometrically grounded view of the delivery network, with each part solved by the Algorithms II technique that fits it.

The river check shows where this straight-line view stops being enough. With one road bridge along 27.5 km of the Hooghly, a van's real trip to a cross-river hub is more than twice the straight line. Knowing this is the first step towards distance models that follow the real geography.

Appendix A. Running the Project

The complete source code, data and tests are in the GitHub repository github.com/anipriyo/smart-delivery-network. The project uses Python 3.12.

```
git clone https://github.com/anipriyo/smart-delivery-network.git
cd smart-delivery-network
pip install -r requirements.txt
streamlit run app.py
```

The dashboard then opens in the browser on port 8501 of localhost. The OpenStreetMap extract is part of the repository (`data/2026-10/`), so the algorithms need no download, and only the background street-map tiles need internet access.

Task	Command
Run the 36 tests	`python -m pytest tests/ -v`
Download the OSM extract again (current month)	`python scripts/fetch_osm.py`
Print every number used in this report	`python report/worked_example.py`
Retake the screenshots (with the app running)	`python report/take_screenshots.py`
Rebuild this PDF from `report/report.md`	`python report/build_pdf.py`

Repository layout

Path	Contents
`app.py` , `ui/`	The Streamlit dashboard, with its pages, deck.gl map layers and styling
`src/`	The algorithms (`convex_hull.py` , `closest_pair.py` , `kdtree.py` , `voronoi.py` , `set_cover.py`), the river model (`river.py`), geography and sampling (`geo.py` , `data.py`), and the Walkthrough traces (`traces.py`)
`tests/`	`test_algorithms.py` , `test_geo.py` , `test_river.py`
`scripts/fetch_osm.py`	OpenStreetMap download and cleaning
`data/`	The October 2026 extract and hand-added ferries
`report/`	This report, its screenshots and the scripts that produce them
`Smart_Delivery_Network_Overview_Document.pdf`	The project overview the work started from